



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology
Department of Artificial Intelligence and Data Science

Subject Name	Theory of Computations	A.Y	2026-27
Subject Code	303105306	Semester	5

Assignment-2

Sr No	Question	COs	B.T	Competence
1	Define regular expressions and regular languages with suitable examples. Explain the importance of regular expressions in pattern matching and compiler design.	2	1	Remember
2	Explain the working of Deterministic Finite Automata (DFA) with a suitable state diagram. Describe the equivalence between DFA and Regular Expressions.	2	2	Understand
3	Construct a DFA that accepts all binary strings ending with "01". Test the DFA for the strings: 101, 1101, 1110, and 1001.	2	3	Apply
4	Compare Moore Machine and Mealy Machine based on output generation, state representation, and applications. Convert a given Mealy Machine into an equivalent Moore Machine.	2	4	Analyze
5	Evaluate whether the language $L = \{a^n b^n \mid n \geq 0\}$ is regular or not using the Pumping Lemma for Regular Languages. Justify each step clearly.	2	5	Evaluate
6	Design and minimize a finite automaton for a language of your choice over the alphabet $\{a,b\}$. Also write the equivalent Regular Grammar and explain the equivalence between finite automata and regular grammars.	2, 3	6	Create